

A Standardized Benchmark for Skeleton-Based Rehabilitation Assessment Using Deep Learning

Ali Ismail-Fawaz^{1*}, Maxime Devanne^{1*}, Stefano Berretti², Jonathan Weber¹ and Germain Forestier^{1,3}

¹ IRIMAS, Université de Haute-Alsace, France

² MICC, University of Florence, Italy

³ DSAI, Monash University, Australia

Abstract—Automated assessment of human motion plays a vital role in rehabilitation, enabling objective evaluation of patient performance and progress. Unlike general human activity recognition, rehabilitation motion assessment focuses on analyzing the quality of movement within the same action class, requiring the detection of subtle deviations from ideal motion. Recent advances in deep learning and video-based skeleton extraction have opened new possibilities for accessible, scalable motion assessment using affordable devices such as smartphones or webcams. However, the field lacks standardized benchmarks, consistent evaluation protocols, and reproducible methodologies, limiting progress and comparability across studies. In this work, we address these gaps by first aggregating existing rehabilitation repositories into a unified archive called Rehab-Pile. Second, we propose a general benchmarking framework for robustly and fairly evaluating deep learning methods in this domain. Experimental results show that some approaches offer a good trade-off between performance and efficiency. All datasets and implementations are publicly released to the community to support transparency, reproducibility and evolution: <https://msd-irimas.github.io/pages/DeepRehabPile/>.

I. INTRODUCTION

The assessment and analysis of human motion are critical in many fields, particularly in healthcare and rehabilitation [18]. With advances in computer vision, motion sensors, and artificial intelligence, there is a growing opportunity to enhance rehabilitation therapy through objective and automated motion evaluation. This is especially important for monitoring patient progress, tailoring treatments, and ensuring the correct execution of prescribed exercises.

Rehabilitation motion assessment [20] (see Figure 1), however, differs significantly from other human motion analysis tasks, such as action recognition. While action recognition focuses on classifying motion types (e.g., walking vs. running), rehabilitation assessment requires evaluating the quality and performance of movements within the same class of activity (e.g., determine whether a squat is performed correctly and quantify the deviation from an ideal motion).

This work particularly focuses on skeleton data extracted from video streams, whether RGB or RGB-D. Skeleton data represent the human body as a set of key joint coordinates, offering a compact and interpretable representation of body movement over time. Unlike motion capture (MoCap) systems, which are highly accurate and robust but costly, in-

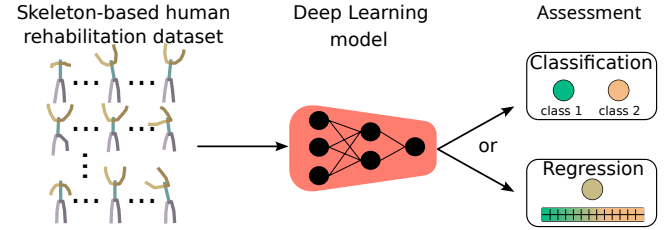


Fig. 1: Illustration of the task of rehabilitation assessment.

trusive, and often impractical for large-scale or at-home use, video-based approaches can be implemented with affordable, readily available devices such as smartphones, webcams, or depth cameras. This accessibility aligns with the goal of making automated rehabilitation assessment widely available to the general public. Deep learning methods, with their ability to model complex spatio-temporal patterns in skeleton data, have emerged as a powerful tool for addressing the challenges of rehabilitation motion assessment [12], [23].

However, many approaches have been evaluated on a very limited number of datasets. The reliance on a small set of benchmarks raises concerns about the generalizability of the methods. Hence, it remains unclear how well these approaches perform across diverse populations, exercise types, or real-world settings. In addition, even when the same datasets are used for evaluation, inconsistencies in experimental protocols across different work make it difficult to ensure fair comparisons. Differences in data splits, pre-processing steps, or evaluation metrics are often not clearly reported, leading to ambiguity in reported performance and limiting the ability to draw reliable conclusions.

Finally, reproducibility remains a significant challenge in this domain. The code and implementation details of many proposed methods are not fully released, which complicates independent validation of results and slows the adoption of effective techniques in practical applications.

To address these limitations, this work makes the following key contributions:

- We aggregate existing skeleton-based rehabilitation motion repositories into a single dedicated archive containing 60 datasets, called Rehab-Pile;
- We benchmark various deep learning approaches for rehabilitation assessment through a unified framework ensuring robust and fair comparisons;

*These authors contributed equally

- To promote reproducibility, we publicly release our aggregated Rehab-Pile and all re-implemented codes.

By addressing these challenges, this paper aims to advance the state-of-the-art in rehabilitation motion assessment, posing a reliable foundation for future research and applications in personalized and scalable rehabilitation solutions.

II. RELATED WORK

In recent years, many deep learning approaches have been proposed for skeleton-based rehabilitation assessment. Inspired by early work on human action recognition, the first deep learning models for rehabilitation assessment were mainly based on Convolutional Neural Networks (CNNs) [12], [30] or Recurrent Neural Networks (RNNs) [19], [41]. To better exploit the inherent structure of skeleton data, several graph-based approaches have subsequently been explored. Notably, the Spatio-Temporal Graph Convolutional Network (STGCN) [5] combines spatial feature learning through Graph Convolutional Networks (GCNs) with temporal modeling via Long Short-Term Memory (LSTM) networks. The success of this model has motivated further extensions, including the incorporation of attention mechanisms [31] and ensemble-based approaches [40]. More recently, Transformer architectures have also been adopted, for instance for spatio-temporal attention [21] and for masked motion modeling [17].

However, most of these approaches have mainly been evaluated on only two repositories, KIMORE [4] and UIPRMD [34], while several other datasets are publicly available. In addition, only a few works have publicly released their code, which complicates reproducibility and fair comparison. In this work, we aim to overcome these limitations by systematically comparing several deep learning approaches for skeleton-based rehabilitation assessment across multiple repositories. While some existing studies have reviewed prior work [20], [36], none of these reviews provides an experimental comparison of methods across multiple rehabilitation datasets. We believe that our benchmark can serve as a solid, reproducible, and scalable basis for evaluating and comparing rehabilitation assessment approaches in a fair and robust manner.

III. PROPOSED BENCHMARK

A. Skeleton sequence representation

In this work, we aim to analyse rehabilitation motion represented as skeleton sequences extracted from videos. A skeleton-based human motion sequence S is a sequence of L skeletons, where each skeleton is represented by a set of J human joints in D dimensions. Such a sequence can be represented as a multivariate time series of $J \cdot D$ variables, each of length L , as illustrated in Figure 2.

B. Deep Learning models for Rehabilitation Assessment

Some of the deep learning models used in our benchmark were originally proposed for Time Series Classification (TSC) in both univariate and multivariate cases, while others were specifically designed for human motion datasets,

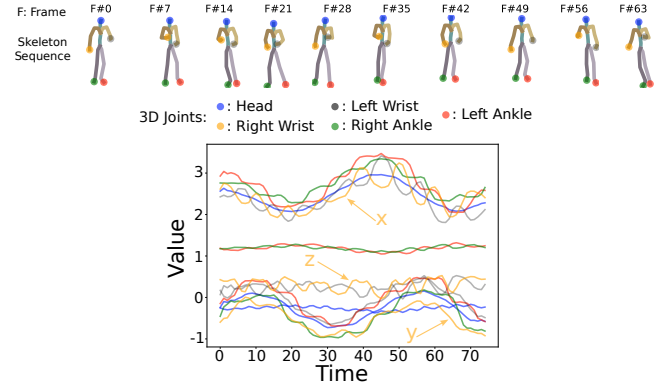


Fig. 2: Skeleton sequence representation into a multivariate time series.

such as those used in activity recognition and rehabilitation assessment. The considered models can be divided into four groups: (1) convolution-based, (2) recurrent-based, (3) transformer-based, and (4) graph-based.

1) *Hybrid Inception (H-Inception)*: Initially proposed for image classification, Inception [33] is a convolution-based deep learning model. It mainly differentiates itself from other convolution-based architectures through a multiplexing convolution approach, which applies multiple convolutional layers with different characteristics to the same input, capturing patterns at various scales.

In 2020, Inception was adapted for TSC [15] by replacing two-dimensional convolutions with one-dimensional ones, making it suitable for processing time series data. In 2022, Ismail-Fawaz et al. [14] introduced a set of hand-crafted, non-trainable convolutional filters designed to detect increasing trends, decreasing trends, and peaks in time series data. To incorporate these features, the Inception architecture was adapted by integrating these hand-crafted filters into the first layer. This modified architecture, known as Hybrid Inception (H-Inception), ends with a global average pooling operation summarizing extracted features before being processed for the downstream task.

2) *LITEMV*: Given the high number of parameters in models such as H-Inception, a lighter architecture, Light Inception with boosting techniques for MultiVariate data (LITEMV), has been proposed in [13]. Leveraging boosting techniques such as multiplexing convolution, hand-crafted convolutional filters, dilated convolutions and depthwise separable convolutions, the LITEMV model results in a significant reduction in the number of parameters, while maintaining the performance of state-of-the-art models for time series classification. The LITEMV architecture, detailed in Figure 3, consists of three convolutional blocks. The final block's output undergoes a global average pooling operation before being fed into the downstream task layer.

3) *Motion Gated Recurrent Unit (MotionGRU)*: MotionGRU was first proposed in [38] for motion prediction. It has since been applied in various work for human motion recognition [9] or generation [11]. The model utilizes GRU layers on raw input sequences to capture global dynamics

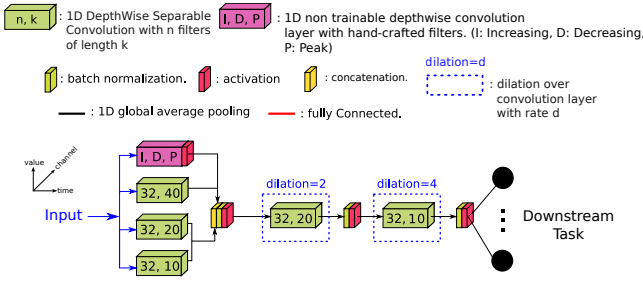


Fig. 3: The Light Inception with boosting Techniques Multi-Variate (LITEMV) [13] deep learning architecture, originally proposed for multivariate TSC.

and summarize sequence patterns into a single vector. The architecture consists of two GRU layers followed by a fully connected layer with an activation function to reduce dimensionality before being fed into the downstream task layer (more architecture details are presented in Figure 4).

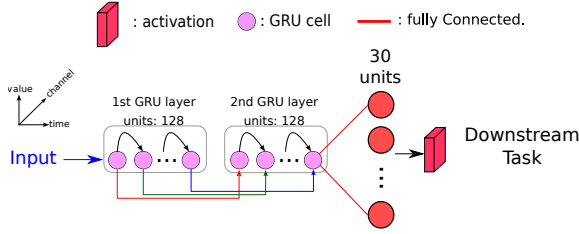


Fig. 4: MotionGRU [38] adapted for skeleton-based rehabilitation assessment.

4) *Convolution Long Short-Term Memory (ConvLSTM)*: Originally proposed in [27] for the action recognition task using wearable sensor activity, the ConvLSTM network combines convolution operations that detect local patterns and LSTM recurrent layers that capture global temporal dependencies. This architecture first applies four convolutional blocks, each consisting of a 2D convolution operation with a kernel width of 1, followed by an activation function. The output of the fourth convolutional block is then passed through two LSTM layers and dropout operations before being fed into the final downstream task.

5) *Vanilla Transformer (VanTran)*: Since its introduction in Transformers for text translation [35], the self-attention mechanism has been widely used for human motion sequences in various tasks such as generation [29], recognition [6] and rehabilitation assessment [21]. In our work, we retain the encoder architecture of [29] and adapt it for classification or regression tasks.

A detailed view of the VanTran architecture is shown in Figure 5. First, the input multivariate sequence is embedded using a bottleneck layer, implemented as a one-dimensional convolution with a unit kernel size. Next, four attention blocks are used. Each attention block consists of a multi-head self-attention mechanism, followed by a residual connection and a Feed-Forward Neural Network (FFNN). Finally, a Global Average Pooling operation is applied before passing the output to the downstream task.

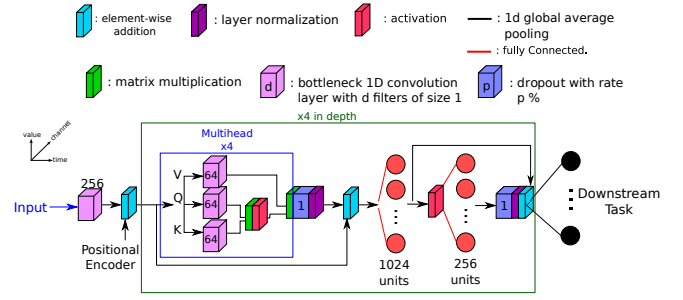


Fig. 5: The Vanilla Transformer (VanTran) architecture inspired from [29].

6) *Convolution Transformer (ConvTran)*: In time series analysis, the Transformer architecture has been recently combined with convolutions in [7]. The ConvTran architecture leverages the 1+1D convolutional embedding technique introduced in [8]. Furthermore, ConvTran utilizes efficient Relative Positional Encoding (eRPE) to enhance the self-attention mechanism by encoding positional relationships directly into the attention computation. As a result, the architecture includes only a single attention block, with no residual connections, followed by a FFNN before the output is passed to the downstream task.

7) *Spatio-Temporal Graph Convolution Network (STGCN)*: Due to the nature of skeleton data, GCNs [16] have been widely used for human motion analysis such as activity recognition [39] and rehabilitation assessment [5]. In this work, we follow [5] and consider three GCN blocks in cascade. Each block consists of 2D convolution layers followed by graph convolution layers that capture the correlation between skeleton joints. Each GCN block is enhanced with an attention mechanism to learn importance weights for each joint. The extracted features are then aggregated using multiple LSTM layers before being fed to the downstream task (see Figure 6 for a detailed view of the STGCN architecture and its enhanced version).

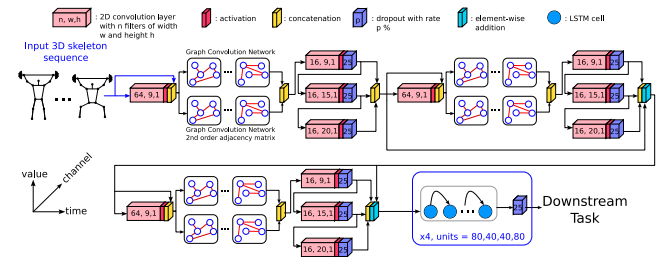


Fig. 6: The Spatio-Temporal Graph Convolution Network (STGCN) [5] originally proposed for skeleton-based rehabilitation assessment.

8) *Downstream Task Layers*: All the models mentioned above produce, for each input sample S , a vector representation h of size D_h that encodes the relevant features. We use these representations for both classification and extrinsic regression tasks.

For the classification task, we add a Fully Connected (FC) layer with softmax activation that transforms the input vector

into a probability distribution. Similarly, for the extrinsic regression task, we apply a FC layer to the output vector representation h . In this case, the output dimension of the FC layer is set to 1, as the goal is to predict a single continuous value. No activation function is applied after the FC layer.

C. The Rehab-Pile

In order to robustly evaluate deep learning models for rehabilitation assessment, we propose the creation of an archive dedicated to the analysis of physical exercises in a medical context, from skeleton sequences. Our archive, Rehab-Pile, is thus derived from 9 publicly available repositories. For each repository, the associated motion analysis task is either binary or multiclass classification, extrinsic regression, or both. Since the analysis is carried out per exercise, we consider each exercise as a separate dataset. Thus, our Rehab-Pile archive contains a total of 60 datasets, 21 for an extrinsic regression and 39 for a classification task.

D. Regression task

1) *EHE_REG*: The *Elderly Home Exercise (EHE)* [2] repository includes six exercises performed daily by people in a nursing home under the guidance of caregivers. The movements of 25 people were captured using a Kinect camera. Each sequence is associated with a label corresponding to the severity level of Alzheimer's disease. In this work, we consider the problem of estimating disease severity as an extrinsic regression task. Therefore, our Rehab-Pile contains six datasets from the original EHE repository.

2) *KIMORE_REG*: The original *KInematic assessment of MOvement for remote monitoring of physical REhabilitation (KIMORE)* repository, presented in [4], comprises 71 subjects, including 40 healthy persons and 31 unhealthy patients performing five different exercises. For each exercise, a subject performed multiple trials during a single session, captured using a Kinect sensor. An expert evaluated their performance and assigned them a continuous score between 0 and 100. Thus, our archive includes five regression datasets from KIMORE repository.

3) *UI-PRMD_REG*: The *University of Idaho-Physical Rehabilitation Movement Data (UI-PRMD)* repository, introduced in [34], comprises recordings of 10 healthy subjects performing physical exercises, correctly and incorrectly, captured using a Kinect camera as well. We consider the segmented version of the repository, where each skeleton sequence corresponds to a realization of one of the 10 exercises. Each sequence is associated with a continuous label, allowing us to consider an extrinsic regression task. Hence, a total of 10 regression datasets from the UI-PRMD repository are included in our Rehab-Pile.

E. Classification task

1) *IRDS_CLF_BN*: The *IntelliRehabDS (IRDS)* [25] repository contains 9 exercises acquired using a Kinect sensor and performed by 14 healthy subjects and 15 unhealthy subjects. Each sequence is associated with a binary label indicating whether the exercise is correctly performed or not.

Our Rehab-Pile thus contains 9 datasets targeting a binary classification task.

2) *KERAAL_CLF_BN and KERAAL_CLF_MC*: The *KERAAL* repository [26] was recorded as part of a clinical study [1] involving 31 patients with low back pain, aged 18 to 70 years. The repository includes recordings of healthy subjects and 12 rehabilitation patients (unhealthy subjects) performing each of three predefined exercises. Each recording was annotated by a doctor. We follow [26] and consider two classification problems, binary and multi-class. Thus, our archive includes 6 datasets from the KERAAL repository, three for binary and three for multi-class classification.

3) *KIMORE_CLF_BN*: In addition to the clinical score prediction task (extrinsic regression) associated with the KIMORE repository, we also consider a binary classification problem, as proposed in [13], to evaluate the success level of each exercise. All sequences with a score less than or equal to 50 are labeled as 0 (incorrect), while all sequences with a score greater than 50 are labeled as 1 (correct). This results in five binary classification datasets.

4) *KINECAL_CLF_BN*: Aiming to assess balance disorders, the KINECAL repository [22] contains movement sequences of 90 participants performing 11 different exercises recorded using a Kinect sensor. For each recording, clinical labels are provided, indicating whether the participant is considered as clinically at risk. We use this annotation and thus consider a binary classification problem. However, among the 90 participants, only 6 are considered clinically at risk. For a fair assessment, we only retain exercises performed by at least five participants labeled as clinically at risk. Therefore, only four exercises from the KINECAL repository are finally retained in our archive.

5) *SPHERE_CLF_BN*: The oldest repository in our archive, *Sensor Platform for Healthcare in a Residential Environment (SPHERE) Walking Up Stair (WUS)* [28], contains sequences recorded by a Kinect sensor of 6 participants walking up stairs. The original goal of this repository was to identify whether the exercise was performed correctly or not. We formulate this task as a binary classification problem. Thus, it results in a single dataset included in our Rehab-Pile.

6) *UCDHE_CLF_BN and UCDHE_CLF_MC*: Proposed in [32], the *University College Dublin Human Exercises (UCDHE)* repository includes two exercises performed by 52 and 54 healthy participants, respectively. Skeleton data are captured from videos using OpenPose [3], providing the 2D positions of 8 joints. Each sequence is annotated with a discrete label indicating whether the exercise is correctly performed or the type of error, if any. We consider two variants of this dataset targeting the original multi-class classification task, but also a binary classification task, where we merge the error types into a single class label (incorrect). Thus, 4 datasets are considered in our Rehab-Pile.

7) *UI-PRMD_CLF_BN*: The original work [34] introducing the UI-PRMD repository also provides, for each sequence, a binary label indicating whether the exercise is correctly performed or not. This allows us to consider a binary classification task, resulting in 10 additional datasets.

IV. EXPERIMENTAL SETUP

A. Data Preparation

1) *Data Pre-processing*: For each individual dataset, we resample all sequences to the average length using a Fourier-based resampling method provided in *scipy*¹. Furthermore, to address potential issues related to feature scaling and to ensure stable model convergence, we apply min-max normalization to the input data. Such normalization is performed independently for each feature dimension across all sequences, frames, and joints. To prevent data leakage, normalization parameters (min and max) are computed exclusively on the training set and then applied to normalize both the training and test sets.

2) *Cross-subject protocol*: For a more robust evaluation, we consider a cross-subject protocol with multiple train-test folds per dataset. When the information is provided, we ensure that test sets contain only sequences from unhealthy subjects, and healthy subjects are included only in training sets. We believe this is closer to a real-world scenario in which we are not interested in assessing healthy subjects. Furthermore, if the number of subjects to assess is greater or equal to 10, we employ a 5-fold split. Otherwise, we consider a leave-one-subject-out approach.

To mitigate potential bias in our results, we perform multiple train-test folds on each dataset. For each fold, we generate separate training and testing sets. The train-test split follows a cross-subject protocol, ensuring that no subject appears in both the training and testing sets. When applicable, we further ensure that the test set contains only samples from unhealthy subjects.

B. Evaluation Criteria

1) *Performance Evaluation*: For classification tasks, we used both accuracy and balanced accuracy to account for potential class imbalance in the datasets. For regression tasks, we consider standard metrics such as the Root Mean Squared Error (RMSE) and the Mean Absolute Error (MAE). It is important to note that prior to training, the score labels of the training samples are normalized between 0.0 and 1.0 by dividing them by the maximum value. During inference, as the model's output is not constrained, we bound it using the following operation:

$$\hat{y} = \begin{cases} 0.0 & \text{if } \mathcal{F}_{reg}(\mathbf{S}) < 0.0 \\ 1.0 & \text{if } \mathcal{F}_{reg}(\mathbf{S}) > 1.0 \\ \mathcal{F}_{reg}(\mathbf{S}) & \text{else,} \end{cases} \quad (1)$$

where $\mathcal{F}_{reg}$ is the regression model. The regression metrics are computed after de-normalizing the predicted labels $\hat{y}$ by multiplying them by the maximum value in the corresponding dataset.

¹<https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.resample.html>

2) *Ensemble Method*: Ensemble methods leverage the principle of combining multiple models to improve robustness and generalization. We consider ensembles of five models: (i) for each fold of a dataset, we train the same model five times with different random initializations; (ii) these five models are used to generate predictions on the test set; (iii) predictions are then averaged to produce the ensemble output in order to compute the corresponding evaluation metrics described above.

3) *Efficiency Evaluation*: To assess model efficiency, we consider the following measures: training time, inference time, number of trainable parameters, and Floating-point Operations per second (FLOPs). This allows us to perform a comprehensive comparison, aiming to identify models that are not only accurate but also computationally efficient.

C. Implementation Details

All competing models are implemented within the same framework, using *TensorFlow-Keras*. Each model is trained for 1500 epochs with a batch size of 64. We employ a learning rate scheduling strategy based on *ReduceLROnPlateau*, with a reduction factor of 0.5, a patience of 50 epochs, and a minimum learning rate of 10^{-4} . Model selection is based on training loss, with the best performing model checkpoint used for final evaluation on the test set. The source code of our work is publicly available here: <https://github.com/MSD-IRIMAS/DeepRehabPile>.

To ensure fairness in computational cost comparisons, all experiments are conducted on the same hardware setup. Specifically, we use a single machine equipped with an NVIDIA RTX 4090 GPU (24GB VRAM), an AMD Ryzen 9 7950X 16-core processor, and 64GB of RAM, running Ubuntu 24.04.

V. EXPERIMENTAL RESULTS

A. Performance

In the following, we compare models' performance considering classification and regression tasks separately. For each dataset, the results correspond to averages across all folds. In addition to average metrics, and given the use of multiple datasets, we also employ a summarization method, the Multi-Comparison Matrix (MCM) [10]. Such a MCM is particularly suitable for analyzing and comparing multiple models across multiple datasets. Each cell corresponds to a pairwise comparison of two models and provides three pieces of information when comparing a row model with a column model: (1) the difference in average performance, highlighted by the cell color; (2) the win-tie-loss count, indicating how many datasets a model wins on or loses against the other; and (3) a p -value assessing the statistical significance of the performance difference between the two models, computed using a Wilcoxon Signed-Rank Test [37]. If a p -value is less than 0.05, the difference in performance is considered statistically significant.

	LITEMV 0.7549	VanTran 0.7430	MotionGRU 0.7282	H-Inception 0.7078	ConvTran 0.7061	ConvLSTM 0.6708	STGCN 0.6455
Mean-Accuracy							
LITEMV 0.7549	Mean-Difference $r > c / r = c / r < c$ Wilcoxon p-value	0.0119 22 / 1 / 16 0.4304	0.0267 26 / 1 / 12 0.0300	0.0470 29 / 1 / 9 0.0001	0.0487 33 / 0 / 6 0.0001	0.0840 28 / 0 / 11 0.0003	0.1094 30 / 0 / 9 $\leq 1e-04$
VanTran 0.7430	-0.0119 16 / 1 / 22 0.4304	-	0.0148 21 / 1 / 17 0.2613	0.0351 22 / 0 / 17 0.1711	0.0368 23 / 2 / 14 0.1230	0.0721 23 / 1 / 15 0.0060	0.0975 27 / 0 / 12 0.0002
MotionGRU 0.7282	-0.0267 12 / 1 / 26 0.0300	-0.0148 17 / 1 / 21 0.2613	-	0.0204 20 / 1 / 18 0.3907	0.0220 19 / 0 / 20 0.4063	0.0574 25 / 2 / 12 0.0030	0.0827 27 / 1 / 11 0.0002
H-Inception 0.7078	-0.0470 9 / 1 / 29 0.0001	-0.0351 17 / 0 / 22 0.1711	-0.0204 18 / 1 / 20 0.3907	-	0.0017 17 / 1 / 21 0.9722	0.0370 21 / 1 / 17 0.1336	0.0624 25 / 0 / 14 0.0065
ConvTran 0.7061	-0.0487 6 / 0 / 33 0.0001	-0.0368 14 / 2 / 23 0.1230	-0.0220 20 / 0 / 19 0.4063	-0.0017 21 / 1 / 17 0.9722	-	0.0353 22 / 1 / 16 0.1693	0.0607 26 / 1 / 12 0.0122
ConvLSTM 0.6708	-0.0840 11 / 0 / 28 0.0003	-0.0721 15 / 1 / 23 0.0060	-0.0574 12 / 2 / 25 0.0030	-0.0370 17 / 1 / 21 0.1336	-0.0353 16 / 1 / 22 0.1693	-	0.0254 20 / 2 / 17 0.2794
STGCN 0.6455	-0.1094 9 / 0 / 30 $\leq 1e-04$	-0.0975 12 / 0 / 27 0.0002	-0.0827 11 / 1 / 27 0.0002	-0.0624 14 / 0 / 25 0.0065	-0.0607 12 / 1 / 26 0.0122	-0.0254 17 / 2 / 20 0.2794	If in bold, then p-value < 0.05

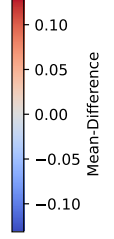


Fig. 7: MCM comparing the seven models on the 39 classification datasets by considering the accuracy metric.

1) *Classification*: Average classification performance across all datasets is reported in Table I, while performance for each individual dataset is detailed in the supplementary material.

TABLE I: Average classification performance of the seven considered models. For both metrics, best results are in **bold** while second-best are underlined.

	Accuracy (%)	Balanced Accuracy (%)
STGCN	64.55 ± 14.91	57.65 ± 10.68
ConvLSTM	67.08 ± 16.37	62.59 ± 13.88
ConvTran	70.61 ± 12.90	65.72 ± 09.69
H-Inception	70.78 ± 12.75	65.67 ± 10.72
MotionGRU	72.82 ± 12.22	<u>67.58 ± 11.88</u>
VanTran	74.30 ± 13.34	67.47 ± 13.79
LITEMV	75.49 ± 11.28	70.34 ± 11.18

We first observe that, for both metrics, the LITEMV model outperforms all others. The second and third best models are MotionGRU and VanTran, with their order inverted depending on the metric. By analyzing the pairwise comparisons between these models considering accuracy in Figure 7, we can nevertheless observe that the difference between LITEMV and VanTran is not statistically significant (p -value greater than 0.05). The same observation holds for MotionGRU and VanTran. This suggests that while the average performance of the LITEMV model across the 39 classification datasets is higher than that of the other models, experiments on additional datasets would be required to draw more definitive conclusions.

In addition, the reported results show that, for both recurrent-based and self-attention-based models, architectures combining convolutions (ConvLSTM and ConvTran) achieve lower average performance than their standard counterparts (MotionGRU and VanTran). This suggests that incorporating additional convolutional features may hinder the models' ability to capture the temporal dynamics of sequences for the classification task.

Finally, we observe that the STGCN model performs

worst among all models in our classification setting. To the best of our knowledge, STGCN has not previously been evaluated on classification tasks and has primarily been applied to regression problems. These results suggest that, within the context of rehabilitation assessment, GCN-based architectures may not be well suited to classification tasks.

2) *Extrinsic regression*: For the regression task, average performances for both RMSE and MAE metrics are reported in Table II. Detailed results per dataset can be found in the supplementary material.

TABLE II: Average regression performances of the seven considered models. For both RMSE and MAE metrics, best results are in **bold** while second-best are underlined.

	RMSE	MAE
STGCN	8.1084 ± 12.1624	6.7389 ± 10.2665
ConvLSTM	7.9354 ± 11.9593	6.5461 ± 10.0711
ConvTran	<u>7.3491 ± 10.8790</u>	<u>6.0729 ± 09.0227</u>
H-Inception	7.9342 ± 11.7514	6.3744 ± 09.3453
MotionGRU	7.8016 ± 11.6802	6.2966 ± 09.5963
VanTran	8.1758 ± 12.5573	6.8688 ± 10.5411
LITEMV	6.4286 ± 09.6731	5.3497 ± 08.1396

First, we can observe that the ranking remains the same for both metrics. Similarly to classification tasks, the LITEMV model outperforms others in regression tasks. The second and third best models for regression tasks are ConvTran and MotionGRU, respectively. In contrast, the pairwise comparisons between LITEMV and other models shown in Figure 8 highlight a statistically significant difference. This suggests that the LITEMV model is more suitable for extrinsic regression tasks in rehabilitation assessment.

Furthermore, we can observe that, unlike the classification tasks, the best self-attention-based model is ConvTran, not VanTran. This shows that, for extrinsic regression, considering convolutional features may improve score prediction. A similar observation can be made for recurrent-based models, as the difference in average RMSE between MotionGRU and ConvLSTM is only 0.15, and the p -value is greater than 0.05.

Finally, the only graph-based model in our comparison,

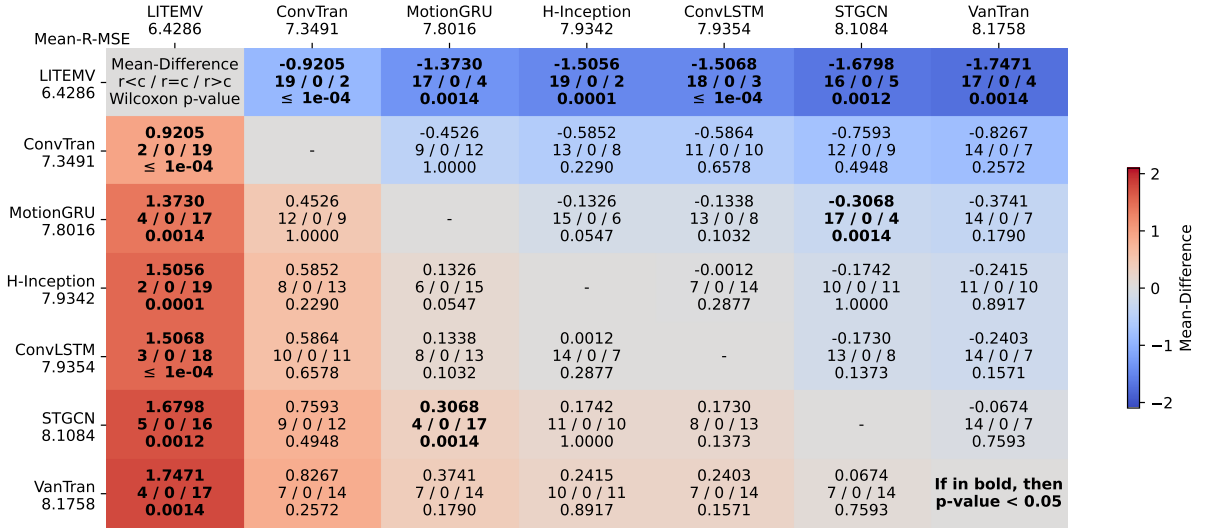


Fig. 8: MCM comparing the seven models on the 21 extrinsic regression datasets by considering the RMSE metric.

STGCN, ranks sixth in terms of both average MAE and RMSE. By examining the pairwise comparisons in Figure 8, we observe that its average RMSE score is very close to those obtained by the H-Inception and ConvLSTM models. Furthermore, no statistically significant differences are observed between these models. This suggests that, in contrast to the classification task, STGCN can be suitable for regression tasks, as also suggested in the literature, although other models appear more appropriate overall. We also note that while the STGCN model is widely used in the human rehabilitation assessment community, it has not previously been evaluated under an experimental setup identical to the one used in this study. In particular, STGCN is often evaluated on either the KIMORE or UI-PRMD repositories. As our experimental protocol differs by including a larger number of datasets and enforcing cross-subject splits, this may explain the observed performance differences. For instance, the detailed results reported in the supplementary material further show that the five datasets on which STGCN performs better than LITEMV belong to either the KIMORE or UI-PRMD repositories.

B. Efficiency

Since the efficiency of the considered models is independent of the downstream task, for comparison, we aggregate the results obtained from all the 60 datasets of our Rehab-Pile. Average results are reported in Table III and presented in detail in the form of strip plots in Figure 9 using *aeon* [24].

First, regarding training time (Figure 9a), we can observe that ConvTran is the fastest model, followed by MotionGRU and LITEMV. Although self-attention-based models are known to be highly dependent on sequence length, only a minimal impact is observed for ConvTran, as it includes a single self-attention layer. Conversely, since VanTran contains multiple self-attention layers, it ranks fifth in terms of training time. Overall, it is important to note that all models exhibit a high standard deviation in training time due to the

TABLE III: Average efficiency results of the seven considered models. For the four efficiency metrics, best results are in **bold** while second-best are underlined.

	Training time (minute)	Inference time (second)	# trainable parameters (Million)	FLOPs (Billion)
STGCN	67.8 ± 99.9	1.2 ± 0.7	0.60 ± 0.90	2.1 ± 2.3
ConvLSTM	<u>02.6 ± 01.9</u>	<u>0.2 ± 0.1</u>	2.70 ± 1.40	2.8 ± 2.9
ConvTran	00.9 ± 00.5	0.5 ± 0.1	<u>0.09 ± 0.04</u>	<u>0.1 ± 0.1</u>
H-Inception	15.5 ± 05.1	0.9 ± 0.2	0.50 ± 0.01	0.3 ± 0.3
MotionGRU	<u>01.2 ± 01.1</u>	0.1 ± 0.0	0.20 ± 0.02	0.1 ± 0.1
VanTran	03.2 ± 03.0	1.5 ± 0.2	6.30 ± 0.01	5.1 ± 6.9
LITEMV	02.0 ± 01.7	0.6 ± 0.2	0.08 ± 0.05	0.1 ± 0.0

varying sequence lengths in our Rehab-Pile. Second, regarding inference time (Figure 9b), MotionGRU and ConvLSTM are the fastest models, which may sound counter-intuitive given their recurrent-based architecture. However, as the average series length in our Rehab-Pile is only 267, it limits the impact of recurrent computation efficiency. Third, concerning the number of trainable parameters (Figure 9c), the two smallest models are LITEMV and ConvTran, with about 1.0 and 1.0, respectively. However, these models appear to be highly dependent on varying dataset characteristics, such as sequence length or the number of joints, as indicated by the large standard deviations. Conversely, models such as H-Inception and VanTran, although much larger, are less sensitive to these characteristics. Finally, in relation to the number of parameters, a similar analysis can be made for the number of FLOPs (Figure 9d), where LITEMV is the most efficient model, followed by ConvTran and MotionGRU. However, only small differences are observed between these models in terms of FLOPs.

C. Trade-off between performance and efficiency

The analysis above shows that none of the seven models demonstrate a clear advantage across all considered metrics. To simultaneously assess performance and efficiency and to highlight the trade-off between these two criteria,

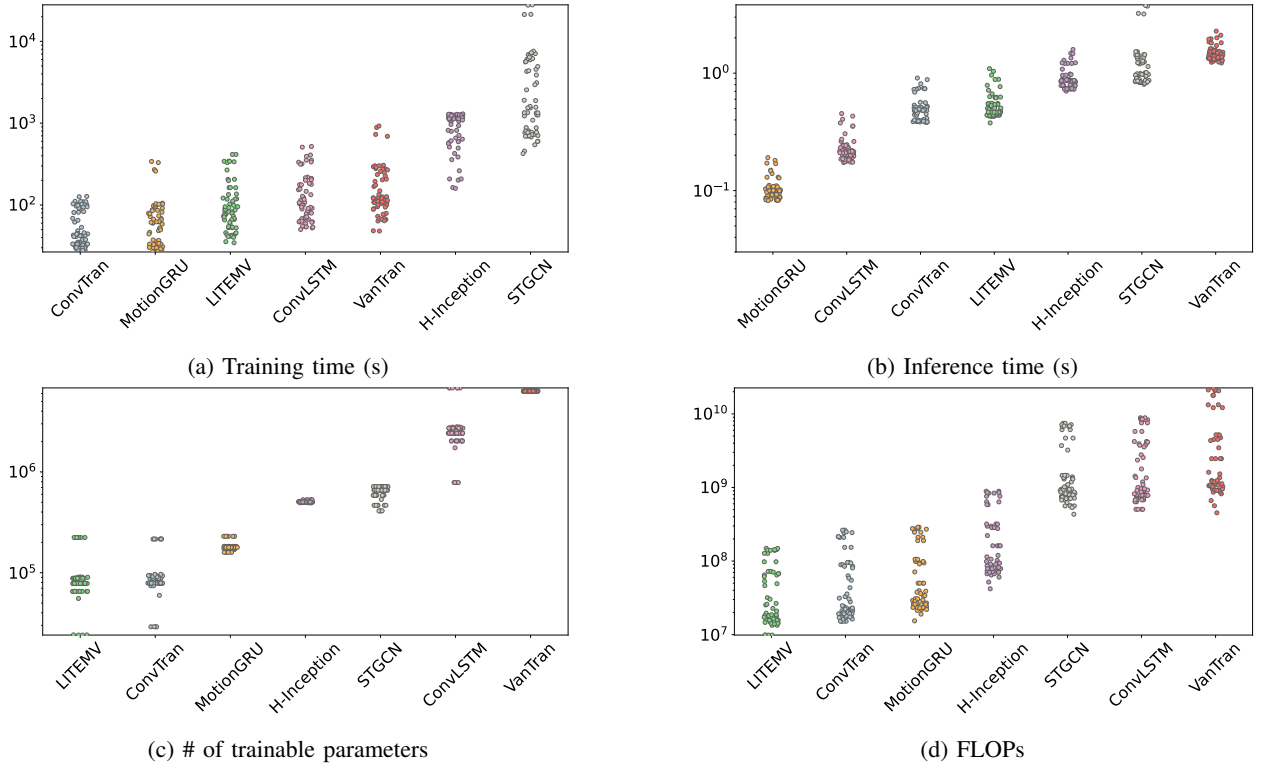


Fig. 9: Efficiency comparison of the considered models on the 60 datasets through four metrics. For each plot, models are ordered by average value in ascending order and values are shown in $\log_{10}$ scale.

independently of the downstream task, we propose a scatter plot in Figure 10. Performance is shown on the y-axis and is measured as the average rank of each model across all 60 datasets in our Rehab-Pile, considering accuracy for classification tasks and MAE for extrinsic regression tasks. Efficiency is represented by the number of FLOPs on the x-axis, while the number of trainable parameters is encoded by the size of the circles.

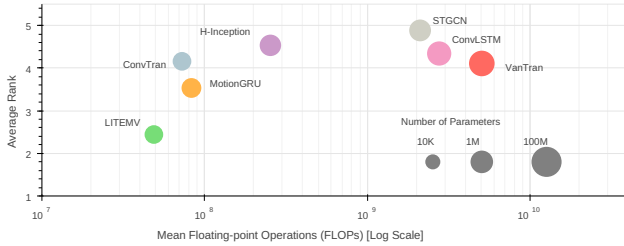


Fig. 10: Trade-off between performance and efficiency of the seven models over the 60 datasets in our Rehab-Pile.

From Figure 10, we can identify three groups. The first group includes more complex models ranked between fourth and fifth on average: STGCN, ConvLSTM, and VanTran. The second group consists of smaller models ranked around fourth in terms of performance: ConvTran, MotionGRU, and H-Inception. Finally, the third group contains a single model, LITEMV, which is the least complex model and achieves the lowest average rank. This suggests that LITEMV offers a favorable trade-off between performance and efficiency for rehabilitation assessment from skeleton data.

VI. CONCLUSIONS

In this work, we address the task of skeleton-based human motion rehabilitation assessment, an important challenge in the healthcare domain. One of the major gaps in this field is the lack of standardized benchmarking, as existing works often use few datasets, inconsistent evaluation protocols, and rarely provide publicly available source code. To address these limitations, we propose a unified framework that evaluates seven deep learning models specifically designed for sequential skeleton data. We benchmark these models across 60 rehabilitation assessment datasets, including 39 for classification and 21 for extrinsic regression. Our experimental results show that the LITEMV model, a convolution-based approach, offers a strong trade-off between performance and efficiency. We believe this work provides a valuable foundation for future research by offering a comprehensive benchmark, complemented by publicly available source code and datasets, to support reproducibility and transparency. Furthermore, our benchmark can serve as a reference for evaluating future methods targeting skeleton-based rehabilitation assessment.

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REFERENCES

- [1] A. Blanchard, S. M. Nguyen, M. Devanne, M. Simonnet, M. L. Goff-Pronost, and O. Rémy-Néris. Technical feasibility of supervision of stretching exercises by a humanoid robot coach for chronic low back pain: The r-cool randomized trial. *BioMed Research International*, 2022:1–10, mar 2022.
- [2] X. Bruce, Y. Liu, K. C. Chan, Q. Yang, and X. Wang. Skeleton-based human action evaluation using graph convolutional network for monitoring alzheimer's progression. *Pattern Recognition*, 119:108095, 2021.
- [3] Z. Cao, G. Hidalgo Martinez, T. Simon, S. Wei, and Y. A. Sheikh. Openpose: Realtime multi-person 2d pose estimation using part affinity fields. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2019.
- [4] M. Capecci, M. G. Ceravolo, F. Ferracuti, S. Iarlori, A. Monteriu, L. Romeo, and F. Verdini. The kimore dataset: Kinematic assessment of movement and clinical scores for remote monitoring of physical rehabilitation. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 27(7):1436–1448, 2019.
- [5] S. Deb, M. F. Islam, S. Rahman, and S. Rahman. Graph convolutional networks for assessment of physical rehabilitation exercises. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 30:410–419, 2022.
- [6] J. Do and M. Kim. Skateformer: skeletal-temporal transformer for human action recognition. In *European Conference on Computer Vision*, pages 401–420. Springer, 2024.
- [7] N. M. Foumani, C. W. Tan, G. I. Webb, and M. Salehi. Improving position encoding of transformers for multivariate time series classification. *Data mining and knowledge discovery*, 38(1):22–48, 2024.
- [8] S. N. M. Foumani, C. W. Tan, and M. Salehi. Disjoint-cnn for multivariate time series classification. In *2021 International Conference on Data Mining Workshops (ICDMW)*, pages 760–769. IEEE, 2021.
- [9] C. Guo, X. Zuo, S. Wang, S. Zou, Q. Sun, A. Deng, M. Gong, and L. Cheng. Action2motion: Conditioned generation of 3d human motions. In *Proceedings of the 28th ACM International Conference on Multimedia*, pages 2021–2029, 2020.
- [10] A. Ismail-Fawaz, A. Dempster, C. W. Tan, M. Herrmann, L. Miller, D. F. Schmidt, S. Berretti, J. Weber, M. Devanne, G. Forestier, et al. An approach to multiple comparison benchmark evaluations that is stable under manipulation of the compare set. *arXiv preprint arXiv:2305.11921*, 2023.
- [11] A. Ismail-Fawaz, M. Devanne, S. Berretti, J. Weber, and G. Forestier. A supervised variational auto-encoder for human motion generation using convolutional neural networks. In *International Conference on Pattern Recognition and Artificial Intelligence*, pages 166–181. Springer, 2024.
- [12] A. Ismail-Fawaz, M. Devanne, S. Berretti, J. Weber, and G. Forestier. Weighted average of human motion sequences for improving rehabilitation assessment. In *International Workshop on Advanced Analytics and Learning on Temporal Data*, pages 131–146. Springer, 2024.
- [13] A. Ismail-Fawaz, M. Devanne, S. Berretti, J. Weber, and G. Forestier. Look into the lite in deep learning for time series classification. *Int. Journal of Data Science and Analytics*, pages 1–21, 2025.
- [14] A. Ismail-Fawaz, M. Devanne, J. Weber, and G. Forestier. Deep learning for time series classification using new hand-crafted convolution filters. In *IEEE Int. Conf. on Big Data*, pages 972–981. IEEE, 2022.
- [15] H. Ismail Fawaz, B. Lucas, G. Forestier, C. Pelletier, D. F. Schmidt, J. Weber, G. I. Webb, L. Idoumghar, P.-A. Muller, and F. Petitjean. Inceptiontime: Finding alexnet for time series classification. *Data Mining and Knowledge Discovery*, 34(6):1936–1962, 2020.
- [16] T. N. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*, 2016.
- [17] I. Kourbane, P. Papadakis, and M. Andries. Ssl-rehab: Assessment of physical rehabilitation exercises through self-supervised learning of 3d skeleton representations. *Computer Vision and Image Understanding*, 251:104275, 2025.
- [18] J. Li, J. Xue, R. Cao, X. Du, S. Mo, K. Ran, and Z. Zhang. Finerehab: A multi-modality and multi-task dataset for rehabilitation analysis. In *International Workshop on Computer Vision in Sports (CVsports) at CVPR 2024*, 2024.
- [19] Y. Liao, A. Vakanski, and M. Xian. A deep learning framework for assessing physical rehabilitation exercises. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 28(2):468–477, 2020.
- [20] Y. Liao, A. Vakanski, M. Xian, D. Paul, and R. Baker. A review of computational approaches for evaluation of rehabilitation exercises. *Computers in biology and medicine*, 119:103687, 2020.
- [21] A. Marusic, S. Mai Nguyen, and A. Tapus. Physioformer: A spatio-temporal transformer for physical rehabilitation assessment. In *International Conference on Social Robotics*, pages 169–179. Springer, 2024.
- [22] S. Maudsley-Barton and M. H. Yap. Kinecal: a dataset for falls-risk assessment and balance impairment analysis. *Scientific data*, 10(1):633, 2023.
- [23] C. Mennella, U. Maniscalco, G. De Pietro, and M. Esposito. A deep learning system to monitor and assess rehabilitation exercises in home-based remote and unsupervised conditions. *Computers in Biology and Medicine*, 166:107485, 2023.
- [24] M. Middlehurst, A. Ismail-Fawaz, A. Guillaume, C. Holder, D. Guijo-Rubio, G. Bulatova, L. Tsaprounis, L. Mentel, M. Walter, P. Schäfer, et al. aeon: a python toolkit for learning from time series. *Journal of Machine Learning Research*, 25(289):1–10, 2024.
- [25] A. Miron, N. Sadawi, W. Ismail, H. Hussain, and C. Grosan. Intellirehabs (irds)—a dataset of physical rehabilitation movements. *Data*, 6(5):46, 2021.
- [26] S. M. Nguyen, M. Devanne, O. Remy-Neris, M. Lempereur, and A. Thepaut. A medical low-back pain physical rehabilitation database for human body movement analysis. In *International Joint Conference on Neural Networks*, 2024.
- [27] F. J. Ordóñez and D. Roggen. Deep convolutional and lstm recurrent neural networks for multimodal wearable activity recognition. *Sensors*, 16(1):115, 2016.
- [28] A. T. Paiement, L. Tao, S. L. Hannuna, M. Camplani, D. Damen, and M. Mirmehdi. Online quality assessment of human movement from skeleton data. In *British Machine Vision Conference*, 2014.
- [29] M. Petrovich, M. J. Black, and G. Varol. Action-conditioned 3d human motion synthesis with transformer vae. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10985–10995, 2021.
- [30] M. J. Raihan, M. A. R. Ahad, and A.-A. Nahid. A deep learning model with interpretable squeeze-and-excitation for automated rehabilitation exercise assessment. *Medical & Biological Engineering & Computing*, pages 1–17, 2025.
- [31] K. Réby, I. Dulau, G. Dubrasquet, and M. B. Aimar. Graph transformer for physical rehabilitation evaluation. In *2023 IEEE 17th International conference on automatic face and gesture recognition (FG)*, pages 1–8. IEEE, 2023.
- [32] A. Singh, A. Bevilacqua, T. B. Aderinola, T. L. Nguyen, D. Whelan, M. O'Reilly, B. Caulfield, and G. Ifrim. An examination of wearable sensors and video data capture for human exercise classification. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pages 312–329. Springer, 2023.
- [33] C. Szegedy, S. Ioffe, V. Vanhoucke, and A. Alemi. Inception-v4, inception-resnet and the impact of residual connections on learning. In *Proceedings of the AAAI conference on artificial intelligence*, volume 31, 2017.
- [34] A. Vakanski, H.-p. Jun, D. Paul, and R. Baker. A data set of human body movements for physical rehabilitation exercises. *Data*, 3(1):2, 2018.
- [35] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- [36] L. Wei, S. J. Wang, et al. Motion tracking of daily living and physical activities in health care: Systematic review from designers' perspective. *JMIR mHealth and uHealth*, 12(1):e46282, 2024.
- [37] F. Wilcoxon. Individual comparisons by ranking methods. In *Breakthroughs in statistics: Methodology and distribution*, pages 196–202. Springer, 1992.
- [38] H. Wu, Z. Yao, J. Wang, and M. Long. Motionrnn: A flexible model for video prediction with spacetime-varying motions. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 15435–15444, 2021.
- [39] S. Yan, Y. Xiong, and D. Lin. Spatial temporal graph convolutional networks for skeleton-based action recognition. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.
- [40] B. Yu, Y. Liu, X. Zhang, G. Chen, and K. Chan. Egcn: An ensemble-based learning framework for exploring effective skeleton-based rehabilitation exercise assessment. In *Int. Joint Conf. Artificial Intelligence, IJCAI-22*, pages 3681–3687, 2022.
- [41] M. Zaher, A. S. Ghoneim, L. Abdelhamid, and A. Atia. Unlocking the potential of rnn and cnn models for accurate rehabilitation exercise classification on multi-datasets. *Multimedia Tools and Applications*, 84(3):1261–1301, 2025.